

Rahul Vishnuvardhana

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Research Interests: *Applied AI · Machine Intelligence · Decision-Making Systems · AI in Organizational & Policy Contexts*

EDUCATION

Northeastern University — Boston, MA

January 2026 – May 2028

Master of Science in Electrical and Computer Engineering — Machine Intelligence

GPA: 3.8–4.0 | Courses: EECE 5644 Machine Learning & Pattern Recognition, EECE 7205 Fundamentals of Computer Engineering

Sri Sairam Engineering College — Chennai, India

August 2018 – April 2022

Bachelor of Engineering in Electronics and Communication Engineering | GPA: 3.4/4.0

Published Thesis: ML-based Image Analysis for Grain Quality Assessment — Int'l Journal of Engineering Technology and Management Sciences, Vol. 6, July 2022

RESEARCH EXPERIENCE

LEO API Intelligence — Multi-Horizon LSTM for Banking API Failure Prediction

Present

Independent Research Project — Northeastern University

- Designed a multi-horizon deep learning system to analyze API failure patterns and support operational decision-making, incorporating bidirectional 2-layer LSTM, multi-head attention, Conv1d feature extraction, and Focal Loss ($\gamma=3.0$) across 5 API types ($h=1, h=5, h=15$).
- Engineered and curated a 1.94M-row hybrid dataset integrating 5 real-world Kaggle sources (NAB EC2, Credit Card Fraud, PaySim, Network Anomaly, Web Logs) with 43 features and per-API scaling; achieved ROC-AUC 0.8088, PR-AUC 0.5132.
- Built rigorous evaluation infrastructure — ablation studies, conformal prediction, and agent simulations — to capture model value across iterations and communicate performance insights to technical and non-technical audiences.
- Developed and deployed an interactive FastAPI dashboard to Render.com to visualize model predictions and communicate findings; trained on dual-GPU Kaggle T4×2 with PyTorch DataParallel; full codebase on GitHub.

Efficiency of RNN Baseline vs. Transformer for Robot Trajectory Prediction

February 2026 – April 2026

Course Group Project — Northeastern University

- Built a complete 5-script RNN pipeline on the NCLT dataset achieving ADE 0.1494m, FDE 0.3509m, MSE 0.0214 (early stop epoch 74); resolved NumPy normalization bug and produced a full technical research report.
- Introduced physics-based kinematics into the linear Transformer to improve computational efficiency and predictive accuracy; systematically compared against RNN baseline across evaluation metrics.

PROFESSIONAL EXPERIENCE

Hexaware Technologies Private Limited — Chennai, India

Software Engineer

March 2023 – September 2025

- Evaluated and advised on cloud infrastructure and document/data retrieval architectures (Azure), comparing tools and services to support scalable data processing and analytics workflows — directly informing tool adoption decisions.
- Designed and maintained ETL pipelines integrating REST APIs, SFTP feeds, and SQL databases; monitored batch job execution and pipeline health in real time, performing root-cause analysis and resolving failures promptly.
- Authored technical documentation, SOPs, and runbooks translating complex systems into accessible formats for non-technical stakeholders; built structured evaluation frameworks for new tool decisions.

Associate Software Engineer

March 2022 – February 2023

- Developed and tested API integrations and data ingestion workflows, validating payload accuracy and enforcing data quality checks; rapidly self-taught new tooling as project requirements evolved.
- Documented end-to-end data workflows, maintained organized process records, and communicated resolution steps clearly to cross-functional teams.

LEADERSHIP & ACTIVITIES

Rotaract Club Member — Sri Sairam Engineering College, Chennai, India

2018 – 2022

- Led volunteer teams across community service initiatives including blood donation camps, coordinating multi-stakeholder logistics and participant communication for inter-college cultural events.
- Served as primary point of contact for event planning and on-ground execution, developing structured coordination and post-event documentation practices transferable to research project support.

TECHNICAL SKILLS

AI / ML: PyTorch, Multi-Horizon LSTM, Multi-Head Attention, Focal Loss, XGBoost, Scikit-learn, LLMs, RAG (learning)

Data Science: Python, Pandas, NumPy, SQL, R, Matplotlib, Seaborn, Ablation Studies, Conformal Prediction

Front-End / Viz: FastAPI, REST APIs, Render.com deployment, Dashboard design, HTML/CSS (basic)

Cloud & Tools: Microsoft Azure (AZ-900 • DP-203 certified), Git/GitHub, Linux/Unix, SQLite, Kafka, Redis, AutoSys, ODI

Microsoft 365: Excel (pivot tables, KPI reporting), Word, PowerPoint, Outlook